Simulation
IEORE4404
Fall 2025
Calvin Chang Homework 4

Problem 1.

Solution:

a) F is the conditional distribution of $Y \sim G$ given that $a \leq Y \leq b$

$$\begin{aligned} P(Y \leq a|a \leq Y \leq b) &= \frac{P(a \leq Y \leq x)}{P(a \leq Y \leq b)} &(1)\\ &= \frac{G(x) - G(a)}{G(b) - G(a)} &(2)\\ &= F(x) &(3)\end{aligned}$$

b) To generate X using acceptance-rejection method:

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    set Y ~ G
    if a <= Y <= b:
        X = Y (accept)
    else:
        (reject)
    Output X
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The accepted sample will then have the distribution of:
$P(X \leq x) = P(Y \leq x|a \leq x \leq b) = F(x)$

Problem 2.

Solution:
Proposal distribution. start with choosing a proposal g:

$$g(x, y) = \frac{1}{2\pi} exp(-\frac{x^2 + y^2}{2})$$

Bounding the ratio.

$$\frac{f^*(x, y)}{g(x, y)} = 2\pi \exp\left(-\frac{1}{2}(x^2 + y^2) + x\sin(xy)\right).$$

Since $\sin(xy) \in [-1, 1]$, it can be simplified to:

$$-\frac{1}{2}(x^2 + y^2) + x\sin(xy) \leq -\frac{1}{2}x^2 + |x| \leq \max_{x \in \mathbb{R}} \left(-\frac{1}{2}x^2 + |x|\right) = \frac{1}{2},$$

where the maximum occurs at $x = \pm 1$ and $y = 0$. Therefore we have,

$$\frac{f^*(x, y)}{g(x, y)} \leq 2\pi e^{1/2} =: M'.$$

Acceptance–rejection algorithm. The procedure is:

1. Generate $(X,Y) \sim g(x,y)$, i.e. $X,Y \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$.
2. Generate $U \sim \text{Uniform}(0,1)$.
3. Accept (X,Y) if
$$U \leq \frac{f^*(X,Y)}{M'g(X,Y)} = \exp\left(-\frac{1}{2}(X^2+Y^2) + X\sin(XY) - \frac{1}{2}\right),$$
otherwise reject and repeat.
4. Output X

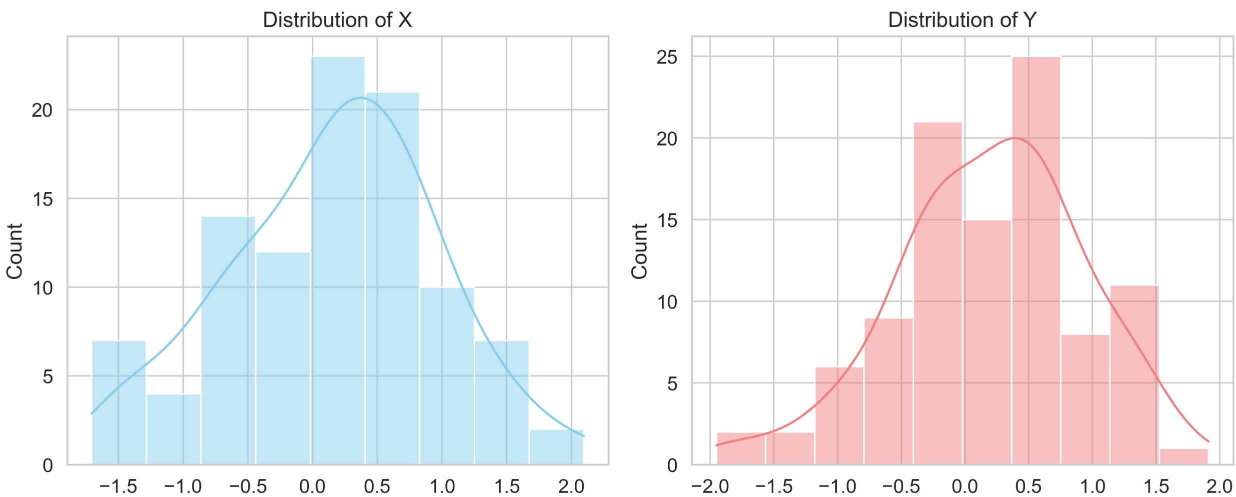

Figure 1: distributions of X and Y

Problem 3.

Solution:
Understanding the region For a given $x \in [-1,1]$,
$$-|x|^{-1/2} \ \leq \ y \ \leq \ |x|^{-1/2}.$$
As $x \to 0$, the allowed range for y grows without bound, but the overall area is finite since
$$\text{area}(A) = \int_{-1}^{1} 2|x|^{-1/2}\,dx = 2\int_0^1 2x^{-1/2}\,dx = 8.$$

Rejection sampling procedure

1. Generate (X,Y) uniformly on the bounding box $[-1,1]\times[-M,M]$:
$$U_1, U_2 \overset{\text{iid}}{\sim} \text{Unif}(0,1) \tag{4}$$
$$X = 2U_1 - 1 \in [-1,1] \tag{5}$$
$$Y = M(2U_2 - 1) \in [-M,M] \tag{6}$$

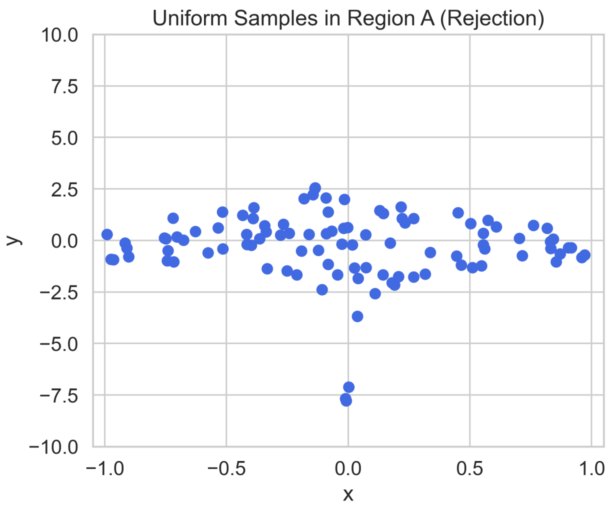

Figure 2: 100 pairs of X Y

2. If $|Y| \le \min\{|X|^{-1/2},\, M\}$, *accept* (X, Y). otherwise *reject*
3. Output (X, Y).

Alternative approach: Direct sampling, no rejection Because a uniform draw on A has cross–section length $2|x|^{-1/2}$ at each x, the marginal density of X on $[-1, 1]$ is

$$f_X(x) = \frac{2|x|^{-1/2}}{\text{area}(A)} = \frac{1}{4}|x|^{-1/2}.$$

By symmetry, $X = \text{sign} \cdot U^2$ where sign $\in \{-1, +1\}$ is Bernoulli$(1/2)$ and $U \sim \text{Unif}(0, 1)$. Conditional on $X = x$, Y is uniform on the vertical segment:

$$Y \mid X = x \;\sim\; \text{Unif}\big[-|x|^{-1/2},\, |x|^{-1/2}\big]\,.$$

1. Draw $B \sim \text{Bernoulli}(1/2)$, $U, V \sim \text{Unif}(0, 1)$ independently.
2. Set $X = (2B - 1)\, U^2 \in [-1, 1]$.
3. Set $Y = (2V - 1)\, |X|^{-1/2}$.
4. Output (X, Y)

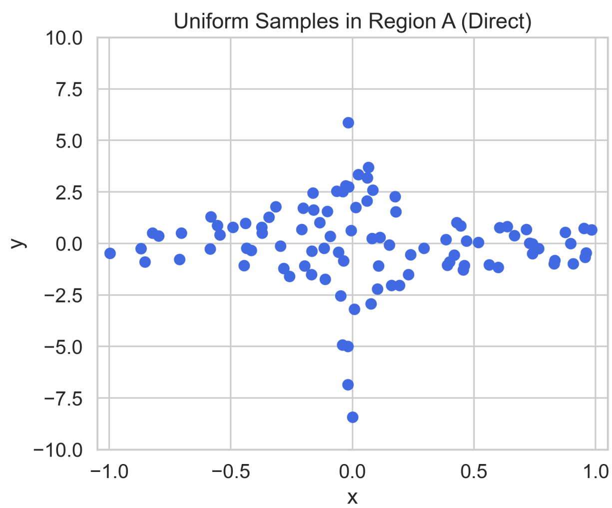

Figure 3: 100 pairs of X Y

Problem 4.

Solution:
For $j \geq 1$,

$$\mathbb{P}(X = j) = \left(\frac{1}{2}\right)^{j+1} + \left(\frac{1}{2}\right)\frac{2^{j-1}}{3^j} = \underbrace{\frac{1}{2}\left(\frac{1}{2}\right)^j}_{\text{(A)}} + \underbrace{\frac{1}{2}\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)^j}_{\text{(B)}}.$$

Note that

$$\sum_{j=1}^{\infty}\left(\frac{1}{2}\right)^j = 1, \qquad \sum_{j=1}^{\infty}\left(\frac{2}{3}\right)^j = 2,$$

so the two terms can be written as a *mixture* with equal weights:

$$\mathbb{P}(X = j) = \frac{1}{2}\,q_1(j) + \frac{1}{2}\,q_2(j),$$

where

$$q_1(j) = \left(\frac{1}{2}\right)^j \quad \text{and} \quad q_2(j) = \frac{1}{2}\left(\frac{2}{3}\right)^j = \frac{1}{3}\left(\frac{2}{3}\right)^{j-1}, \qquad j = 1, 2, \ldots$$

Thus q_1 is the geometric distribution on $\{1, 2, \ldots\}$ with success probability $p = \frac{1}{2}$,

$$\text{Geom}\left(\tfrac{1}{2}\right): \quad \mathbb{P}(J = j) = \left(\tfrac{1}{2}\right)^j,$$

and q_2 is the geometric distribution on $\{1, 2, \ldots\}$ with $p = \frac{1}{3}$,

$$\text{Geom}\left(\tfrac{1}{3}\right): \quad \mathbb{P}(J = j) = \tfrac{1}{3}\left(\tfrac{2}{3}\right)^{j-1}.$$

Composition (mixture) sampling algorithm

1. Draw $B \sim \text{Bernoulli}\left(\frac{1}{2}\right)$ and $U \sim \text{Unif}(0,1)$, independent.
2. If $B = 0$, sample X from $\text{Geom}\left(\frac{1}{2}\right)$ on $\{1, 2, \ldots\}$; e.g.
$$X = \left\lfloor \frac{\log(1-U)}{\log(1-\frac{1}{2})} \right\rfloor + 1 = \left\lfloor \frac{\log(1-U)}{\log(\frac{1}{2})} \right\rfloor + 1.$$
3. If $B = 1$, sample X from $\text{Geom}\left(\frac{1}{3}\right)$ on $\{1, 2, \ldots\}$; e.g.
$$X = \left\lfloor \frac{\log(1-U)}{\log(1-\frac{1}{3})} \right\rfloor + 1 = \left\lfloor \frac{\log(1-U)}{\log(\frac{2}{3})} \right\rfloor + 1.$$
4. Output X

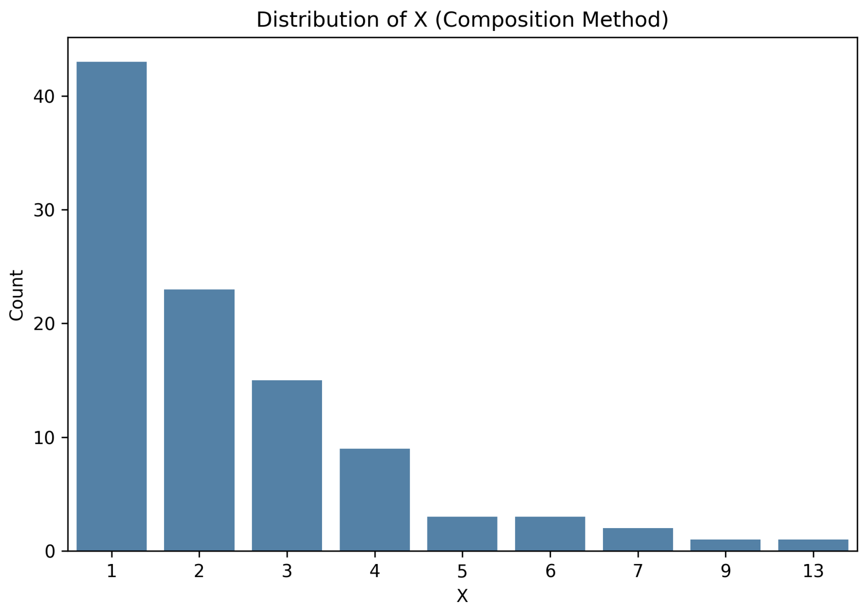

Figure 4: 100 samples of X

Problem 5.

Solution:
For $x \in (0,1]$, write $x^y = e^{y \ln x}$ and compute
$$F(x) = \int_0^\infty e^{y \ln x} e^{-y}\, dy = \int_0^\infty e^{-y(1-\ln x)}\, dy = \frac{1}{1 - \ln x},$$

The cdf is strictly increasing on $(0, 1]$, we can therefore use the inverse–transform method. Let $U \sim \text{Unif}(0, 1)$ and solve $U = F(x)$ for x:

$$U = \frac{1}{1 - \ln x}$$
$$\frac{1}{U} = 1 - \ln x$$
$$\ln x = 1 - \frac{1}{U}$$
$$X = \exp\Big(1 - \frac{1}{U}\Big)$$

Algorithm

1. Generate $U \sim \text{Unif}(0, 1)$.
2. Set $X = \exp\big(1 - \frac{1}{U}\big) \in (0, 1)$.
3. Output X

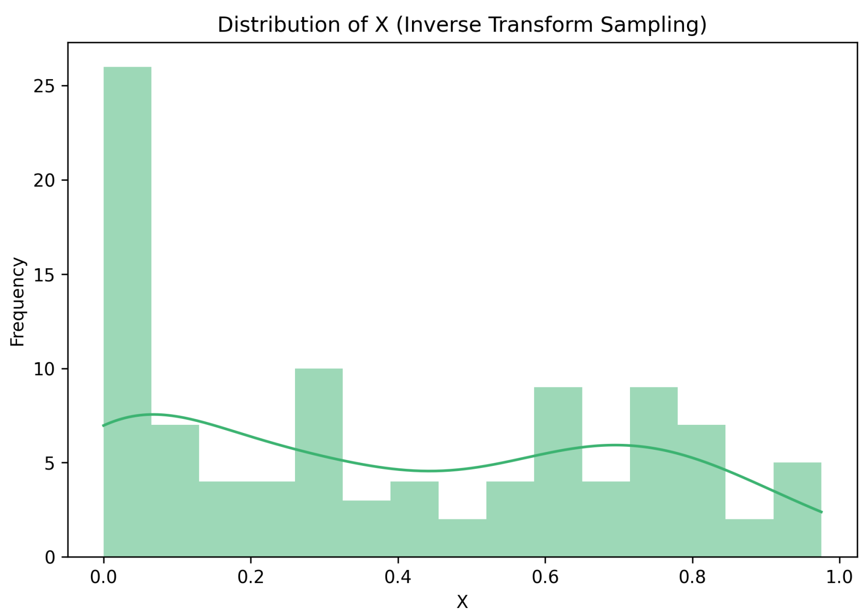

Figure 5: 100 samples of X